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Cypher Analytica — Tableau for Data Visualization and Business Analytics

Level: Beginner to Advanced
Duration: 60 instructor-led hours + 30 hours of independent practice
Suitable for: Students, aspiring data analysts, business analysts, MIS professionals, managers, and business owners
Prerequisites: Basic Excel knowledge and familiarity with tabular data. SQL knowledge is helpful but not mandatory.

Teaching approach: Concept explanation, guided demonstrations, business exercises, dashboard development, and a complete analytics project.

Course objectives

By the end of the course, learners should be able to:

- Connect, organize, and validate data for analysis.
- Build appropriate charts and interactive dashboards.
- Create calculated fields, table calculations, and level-of-detail expressions.
- Use filters, parameters, and actions to support exploration.
- Explain business findings through clear visual storytelling.
- Publish work appropriately and troubleshoot common analytical issues.

Module-wise curriculum

Module	Topics covered	Hours
1. Tableau and Visual Analytics Fundamentals	Business intelligence concepts; Tableau workflow; worksheets, dashboards, and stories; interface navigation; dimensions and measures; discrete and continuous fields; understanding the level of detail in a view	3
2. Data Connectivity and Preparation	Connecting to Excel, CSV, and databases; live connections versus extracts; data types; geographic roles; renaming and hiding fields; splitting fields; pivoting columns; Data Interpreter; NULLs; identifying duplicate records and source-data issues	5

Module	Topics covered	Hours
3. Data Modeling and Combining Data	Table grain; logical and physical layers; relationships versus joins; join types; unions; cardinality and referential integrity concepts; introductory data blending; preventing duplicated measures and validating totals	5
4. Core Visualizations and Formatting	Bar, line, area, scatter, pie, and treemap charts; text tables; highlight tables; heat maps; histograms; box plots; Marks card; labels; tooltips; sorting; groups; hierarchies; aliases; number formatting	6
5. Calculated Fields and Business Logic	Row-level and aggregate calculations; arithmetic; string and date functions; type conversion; Boolean logic; IF, ELSEIF, and CASE; IFNULL and ZN; conditional aggregation; ratios; weighted averages; aggregate versus nonaggregate errors	6
6. Filters, Parameters, Sets, and Order of Operations	Extract, data-source, context, dimension, and measure filters; date filters; Top N filtering; parameter-driven calculations; sets and combined sets; dynamic selection; order of operations; understanding which filters affect a calculation	5
7. Table Calculations and Time-Based Analysis	Quick table calculations; running totals; percentage of total; difference and percentage difference; rank; moving averages; LOOKUP, WINDOW_SUM, and WINDOW_AVG; addressing and partitioning; Compute Using; period comparisons	6
8. Level-of-Detail Expressions	FIXED, INCLUDE, and EXCLUDE; calculation grain versus view grain; customer-level and category-level metrics; first-purchase analysis; comparing individual and group results; interaction with filters; LOD expressions versus table calculations	6
9. Maps and Analytical Features	Symbol and filled maps; geographic hierarchies; unknown locations; latitude and longitude; reference lines and	4

Module	Topics covered	Hours
	bands; distributions; trend lines; introductory forecasting and clustering; interpreting results and limitations	
10. Dashboard Design and Visual Storytelling	Dashboard planning; tiled and floating layouts; containers; spacing and visual hierarchy; KPI displays; filter, highlight, URL, parameter, and set actions; navigation; device layouts; accessible colors; annotations and storytelling	6
11. Publishing, Sharing, and Performance	Workbook and packaged-workbook formats; publishing concepts for Tableau Public, Cloud, and Server; permissions and refresh concepts; appropriate handling of public versus confidential data; reducing unnecessary marks and filters; introductory performance recording	3
12. Capstone Validation and Assessment	Final dashboard refinement; source reconciliation; testing calculations and actions; documentation; insight presentation; practical assessment and viva	5
Total	**Guided instruction, practical work, and assessment**	**60**

Practical exercises

Learning area	Suggested task
Data preparation	Identify duplicate transaction IDs, missing values, and incorrect data types
Data modeling	Combine orders, customers, and products without inflating sales
Chart selection	Choose suitable visuals for comparisons, distributions, trends, and relationships
Calculated fields	Calculate revenue, profit margin, order value, and customer segments
Parameters and sets	Build a selectable metric and a user-controlled Top N product view
Table calculations	Calculate running sales, moving averages, and percentage of total

Learning area	Suggested task
LOD expressions	Calculate customer lifetime sales and compare them with individual orders
Filter behavior	Explain how a context filter changes a Top N or FIXED calculation
Maps	Compare regional performance and resolve unrecognized locations
Dashboard actions	Create a summary dashboard that filters a detailed analysis view
Validation	Reconcile dashboard totals with source data and investigate discrepancies

Capstone project options

1. **Sales Performance Dashboard:** Revenue, profit, margin, regional comparisons, customer segments, product contribution, and trends.
2. **HR Analytics Dashboard:** Headcount, department distribution, hires, exits, tenure, and clearly defined attrition measures.
3. **Retail Customer Analysis:** Purchase frequency, average order value, customer contribution, repeat purchases, and first-purchase cohorts.
4. **Institute Performance Dashboard:** Admissions, course enrollment, batch performance, attendance, and fee collections.
5. **Operations and Delivery Dashboard:** Order volumes, delivery times, delays, geographic patterns, and supplier performance.

Students should develop their project throughout the course, using the final module for validation and presentation.

Mandatory project deliverables

- A documented data source and explanation of its grain.
- A Tableau workbook with organized worksheets and meaningful field names.
- At least eight business calculations, including a table calculation and an LOD expression.
- Two connected dashboards covering executive overview and detailed analysis.
- At least one parameter and two purposeful dashboard actions.

- Clear titles, units, legends, tooltips, and accessible formatting.
- Validation checks for totals, filters, calculations, and interactions.
- A short presentation covering findings, limitations, and proposed actions.
- A published portfolio version using public or synthetic data, where account access is available.

Assessment Structure

Component	Weight
Data preparation and modeling assignments	20%
Calculated fields, table calculations, and LOD assessment	25%
Visualization and dashboard design	20%
Capstone project	25%
Technical viva and troubleshooting	10%
Total	**100%**

Recommended certification criteria: Minimum 60% overall, completion of mandatory assignments, and successful demonstration of a validated capstone dashboard.

Interview preparation

Practice explaining dimensions versus measures, discrete versus continuous fields, relationships versus joins, live connections versus extracts, context filters, order of operations, table calculations versus LOD expressions, duplicate totals, and dashboard-performance issues.